

## **Assignment 2, Poster I: Describing a Spatial Variable**

### **Electricity Access Rate in Africa for 2023**



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John Abboud and Delu Maduekwe  
GEOS 457 - Cartography and Geovisualization  
Instructor: Tigran Melkumyan

Date Submitted: July 31, 2026

# Project Description

## *Purpose and Main Objective*

The purpose of this poster is to describe the distribution of electricity access rates across Africa in 2023. Electricity access rate refers to the percentage of the population that has access to electricity. The project does not test the relationship between two different variables. Instead, it focuses on describing one variable from several geographic and statistical viewpoints.

The poster shows how the understanding of electricity access changes depending on the type of information being examined. This includes comparing the continental level with the state level in Nigeria, comparing percentage rates with the total number of people without electricity, ranking countries with the highest and lowest access rates, and examining the difference between urban and rural areas.

The main message of the poster is that one overall electricity access rate for Africa does not show the large differences across the continent. To describe electricity access accurately, the variable needs to be viewed from several different angles.

## *Data, Methods, and Geography*

Country-level electricity access rates and population data were collected from the World Bank, World Bank Open Data, and ESMAP's *Tracking SDG7: The Energy Progress Report* published in 2024. Information about electricity access at the state level in Nigeria was collected from the USAID Demographic and Health Surveys Program.

Country and state boundary data were obtained from geoBoundaries and are used under the CC BY 4.0 licence (Runfola et al., 2020). The continental choropleth maps use a Quantile classification method. The country-level access rate map received a goodness of variance fit score of 0.965, while the Nigeria state-level map received a score of 0.982.

The continental maps use the Africa Albers Equal Area Conic projection. The Nigeria state-level map uses the WGS 1984 UTM Zone 32N projection. Disputed territories, such as Western Sahara and Abyei, are displayed as "No Data." Their appearance on the maps does not represent an opinion about their political status. Small island countries were not included in the proportional symbol and pie chart maps because the symbols would overlap at the selected map scale.

## *Purpose of Each Graphic Element*

### **Choropleth Map: Population Access to Electricity**

The first choropleth map presents the general continental pattern of electricity access by country. It uses a percentage rate, which allows countries to be compared without the results being strongly influenced by their physical size or total population. A histogram is included with the map to show how electricity access rates are distributed across African countries.

### **Proportional Symbol Map: Population Without Access**

The proportional symbol map shows the total number of people who do not have access to electricity. It is placed beside the access rate choropleth because percentages and total population numbers can present different views of the same issue.

A bar chart also ranks the ten countries with the largest populations without electricity. The chart shows that a small number of highly populated countries represent a large share of Africa's total unelectrified population. Nigeria has the largest electricity access deficit in absolute population terms.

### **Choropleth Map: Nigeria State-Level Access Rate**

The Nigeria map examines electricity access at a smaller geographic scale. It shows differences between individual states that cannot be seen on the continental map. The map reveals a strong north-to-south pattern in electricity access within Nigeria. This demonstrates that national-level values can hide major differences inside a country.

### **Bar Charts: A 96-Percentage-Point Gap**

The two bar charts compare the ten African countries with the highest electricity access rates and the ten countries with the lowest rates. Placing the two groups beside each other makes the large difference between the extremes easier to understand. The charts provide exact values that may be difficult to identify from the colour categories on the choropleth map alone.

### **Proportional Pie Chart Map: Urban and Rural Split**

The proportional pie chart map shows how each country's population without electricity is divided between urban and rural areas. The size of each symbol represents the country's total population.

This map presents another part of the electricity access issue. Some countries may have moderate national access rates while still having a serious lack of electricity in rural areas. This rural gap may not be clear when only the national percentage is examined.

### **Facts and Closing the Gap Sidebars**

The "Facts" and "Closing the Gap" sections provide additional information about electricity access across Africa. They also describe current policy responses, including the World Bank and African Development Bank's Mission 300 initiative. These sections provide useful context without introducing another mapped variable into the poster.

### **One Variable, Five Viewpoints Text Block**

The "One Variable, Five Viewpoints" section explains the main interpretation of the poster. It connects the five main graphic elements and shows why a single map is not enough to fully describe electricity access across Africa.

### ***How the Elements Support the Overall Objective***

Each part of the poster presents a different viewpoint on the same spatial variable. These viewpoints include geographic scale, measurement type, country rankings, and the division between urban and rural populations. The panels are designed to provide different information instead of repeating the same message.

Together, the graphic elements create a more complete understanding of electricity access in Africa. They show where access rates are highest and lowest, where the largest numbers of people remain without electricity, how patterns within Nigeria differ from the continental pattern, and how electricity access differs between urban and rural areas.

Overall, the poster meets the assignment objective by presenting a detailed and multi-viewpoint description of one spatial variable. It demonstrates that electricity access cannot be properly understood through one map or one continental average alone.